

1 Introduction

Options are financial derivatives, meaning that their value is derived from an underlying asset such as a stock. In the equity options market, investors trade standardised option contracts on organised exchanges. These contracts give the holder the right, but not the obligation, to buy or sell a stock at a fixed price within a given time period.

A put option gives the holder the right to sell the underlying stock at a fixed strike price. It is commonly used as a risk management tool when investors are concerned about a possible fall in the stock price. By purchasing a put option, investors can limit their potential losses while still owning the stock. This strategy is known as a protective put and works in a similar way to insurance.

Participants in the options market include individual investors, institutional investors, and professional traders. Risk-averse investors use put options to protect their portfolios, while others may use options to speculate on price movements or to manage risk.

In this project, a protective put strategy is analysed using Tesla Inc. stock as the underlying asset. The analysis focuses on how the use of a put option affects the risk and return characteristics of a stock investment.

2 Black–Scholes Formula

Model assumption. Equity options traded in the US are usually *American*-style, so they may be exercised at any time before expiry. The Black–Scholes model, however, prices *European*-style options, which are exercised only at maturity. In this project, the put option is valued using the Black–Scholes formula and is therefore treated as a European put. As a result, the option premiums reported here are *model-based* values, and any additional value from possible early exercise is not included.

This project examines a risk-management strategy using Tesla Inc. (TSLA) as the underlying asset. The strategy is a *protective put*, which combines a long position in the stock with the purchase of a put option. The put option acts like insurance: it gives the investor the right to sell the stock at a fixed strike price K , which helps to limit losses when the stock price falls.

The theoretical price of a European put option, denoted by P , is given by the Black–Scholes formula:

$$P = Ke^{-rT}N(-d_2) - SN(-d_1), \quad (1)$$

where

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

Here, $N(\cdot)$ denotes the cumulative distribution function of the standard normal distribution. The parameters are:

- S : current stock price (USD),
- K : strike price (USD),
- r : risk-free interest rate (annualised),
- T : time to maturity (years),
- σ : volatility (standard deviation of log returns).

The dataset contains 251 daily observations from 27 December 2023 to 24 December 2024. The main variables used in the analysis are:

- X : Tesla stock price (USD),
- Y_1 : profit from holding the stock only (unhedged) (USD),
- Y_2 : profit from the protective put strategy (hedged) (USD).

Table 1: Key parameters used to compute the hedged profit Y_2

Parameter	Description	Specification
X_0	initial stock price (USD)	261.44
K	put option strike price (USD)	261.44
r	risk-free interest rate	0.04
σ	volatility	0.5
T_t	time to maturity (years)	decreases daily

The time to maturity, T_t , is updated each trading day to reflect the remaining time until the option expires. Hence, T_t decreases over time and approaches zero as maturity is reached.

The hedged profit is computed using a daily mark-to-market approach. On each trading day, the put premium is re-calculated using the Black–Scholes model. Let X_t denote the stock price and P_t the corresponding put premium on day t . The initial values are X_0 and P_0 . The hedged profit is defined by

$$Y_{2,t} = (X_t - X_0) + (P_t - P_0). \quad (2)$$

This definition captures the combined change in the stock value and the option premium relative to their initial levels.

The initial put premium P_0 is computed on the first trading day using the same parameter settings. For this dataset, $P_0 = 45.6326$ USD.

The dataset was randomly partitioned into a 70% training set with 176 samples and a 30% test set with 75 samples.

3 Comparison of Unhedged and Hedged Positions

Unhedged Position (Y_1): The unhedged strategy means holding Tesla stock only. The profit moves one-for-one with the stock price: if the stock falls by \$1, the position loses \$1. There is no downside protection, so losses grow as the stock price decreases.

Hedged Position (Y_2): The hedged strategy is a *protective put*. The investor holds the stock and buys a put option with strike price K , which sets a minimum effective selling price at maturity. At expiry, the put payoff is

$$\max(K - X_T, 0),$$

so if the stock finishes below K , the option offsets part (or all) of the stock loss.

In this project, the hedged position is also tracked *daily* using a mark-to-market approach. The put premium P_t is re-valued each trading day, so the hedged profit changes with both the stock price and the option premium:

$$Y_{2,t} = (X_t - X_0) + (P_t - P_0).$$

This means Y_2 is not only affected by the final payoff at maturity, but also by day-to-day movements in the option premium. As the option approaches maturity, the time to maturity T_t decreases, which generally reduces the option's time value, while the premium remains sensitive to changes in the stock price.

4 Data Analysis

4.1 Data Visualisation and Discussion

A subset of the collected data is presented in Table 2.

Table 2: Subset of Tesla Stock Prices, Profits, and Put Option Premium

Seq	Date	Y_1 (Unhedged)	Y_2 (Hedged)	X (Price)	Put Premium (2 d.p.)
1	2023-12-27	0.00	0.00	261.44	45.63
2	2023-12-28	-8.26	-5.17	253.18	48.72
3	2023-12-29	-12.96	-8.05	248.48	50.54
4	2024-01-02	-13.02	-8.16	248.42	50.49
5	2024-01-03	-22.99	-13.94	238.45	54.68
6	2024-01-04	-23.51	-14.30	237.93	54.84
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
251	2024-12-24	200.84	155.21	462.28	0.00

The dataset was randomly partitioned into a 70% training set with 176 samples and a 30% test set with 75 samples.

Figure 1 shows the scatter plot of unhedged profit (Y_1) against Tesla stock price (X), exhibiting a perfect linear relationship with slope 1.

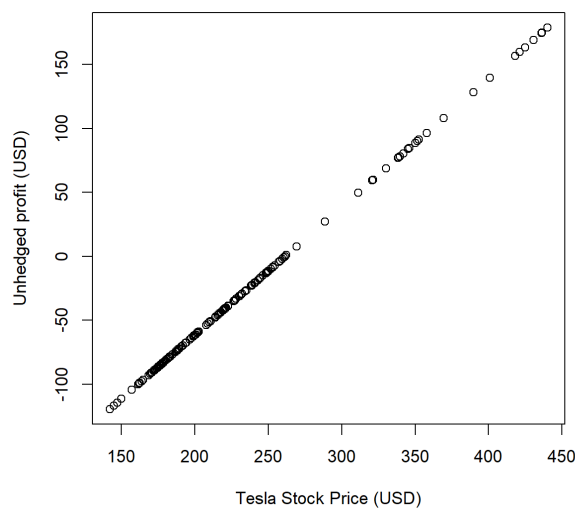


Figure 1: Scatter Plot: Unhedged Profit (Y_1) vs Stock Price (X)

In contrast, Figure 2 shows a nonlinear relationship between hedged profit (Y_2) and stock

price (X), indicating that a simple linear regression is inadequate and a polynomial model can be considered.

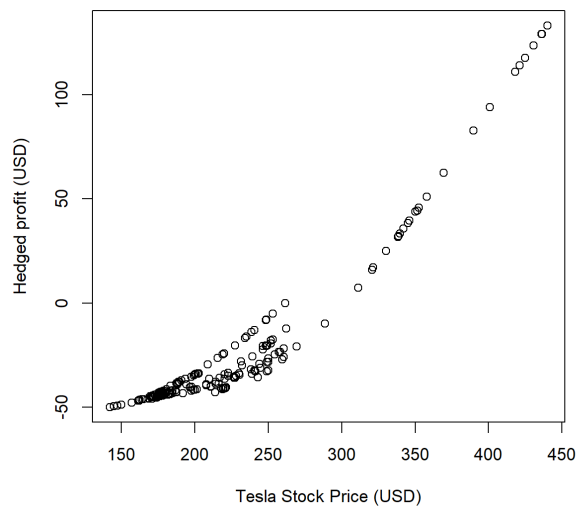


Figure 2: Scatter Plot: Hedged Profit (Y_2) vs Stock Price (X)

Figure 3 shows how the put option premium changes over time. In general, the premium consists of intrinsic value and time value. As the option moves closer to maturity, the remaining time to maturity T_t becomes smaller, so the time value tends to fall. Near the end of the sample, Tesla's stock price is above the strike price K , so the put is out-of-the-money and has zero intrinsic value. With little time left for the stock price to drop below K , the premium falls towards zero as $T_t \rightarrow 0$.

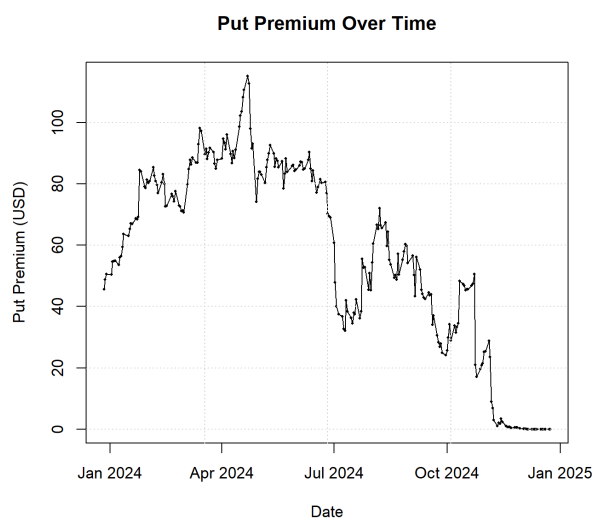


Figure 3: Put Option Premium Over Time

4.2 Model Building and Selection

4.2.1 Data Transformation

Table 3: Variance Inflation Factor Before and After Mean Centering

Model Specification	Variance Inflation Factor
Original Predictor X and X^2	46.70
Centered Predictor X_c and X_c^2	2.48

Table 3 compares multicollinearity before and after mean centring. In the original model, the VIF values are large, indicating severe multicollinearity. After mean centring, the VIF decreases to 2.48, which is below the common threshold of 5. This suggests that mean centring substantially reduces multicollinearity and helps to improve the stability of the coefficient estimates.

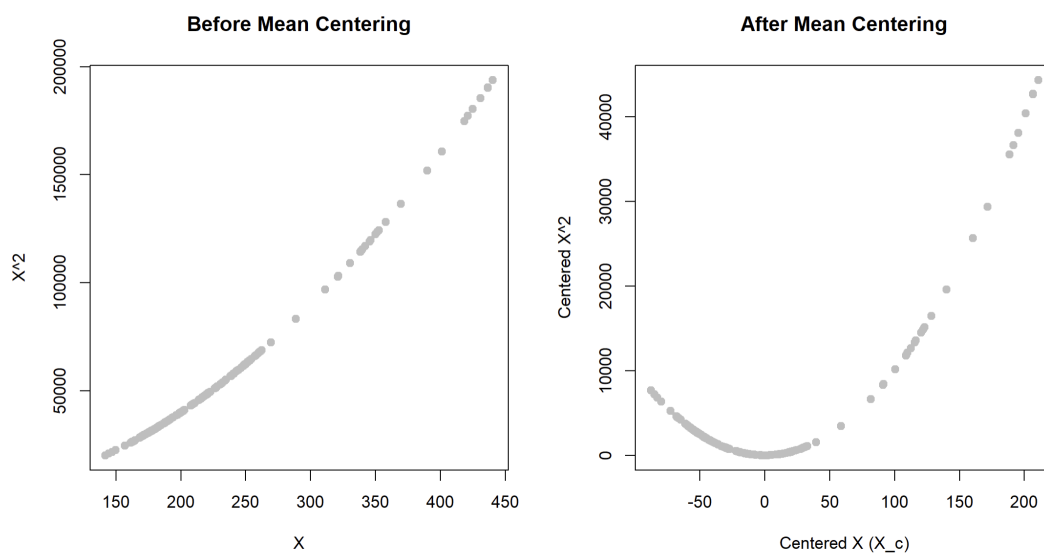


Figure 4: Regressors X and X^2 Before and After Mean Centring

Figure 4 illustrates the relationship between the regressor and its squared term before and after mean centring. Before centring, X and X^2 move closely together, showing a strong association and hence severe multicollinearity. After mean centring, the centred regressor X_c is closer to zero on average, so X_c and X_c^2 are less strongly associated. This weaker dependence is consistent with the substantial reduction in multicollinearity.

4.2.2 Model Forward Selection

Polynomial models of orders 1, 2, and 3 were fitted and evaluated based on term significance, model fit, and residual standard error.

4.2.3 Polynomial Model of Order 1

Table 4: Linear Polynomial Model

Statistic	Value
Adjusted R^2	0.9018
Residual Standard Error σ	12.26
p -value	$< 2 \times 10^{-16}$

For the first-order polynomial model

$$Y = \beta_0 + \beta_1 X_c + \varepsilon,$$

we test whether the regressor has a non-zero effect.

$$H_0 : \beta_1 = 0 \quad \text{and} \quad H_1 : \beta_1 \neq 0.$$

The model is statistically significant with $p < 2 \times 10^{-16}$, so we reject H_0 at the 5% level. This indicates that X_c is associated with Y . However, the adjusted R^2 is 0.9018 and the residual standard error is 12.26, suggesting that the linear specification may be too restrictive and leaves noticeable variation unexplained.

4.2.4 Polynomial Model of Order 2

Table 5: Quadratic Polynomial Model

Statistic	Value
Adjusted R^2	0.9840
Residual Standard Error σ	4.955
Quadratic Term p -value	$< 2 \times 10^{-16}$

For the second-order polynomial model

$$Y = \beta_0 + \beta_1 X_c + \beta_2 X_c^2 + \varepsilon,$$

we test whether the quadratic term has an effect given that X_c is already in the model.

$$H_0 : \beta_2 = 0 \quad \text{and} \quad H_1 : \beta_2 \neq 0.$$

The quadratic term is statistically significant with $p < 2 \times 10^{-16}$, so we reject H_0 at the 5% level. This indicates that X_c^2 is associated with Y given that X_c is already in the model. The adjusted R^2 is 0.9840 and the residual standard error is 4.955, suggesting a much better fit than the first-order specification.

4.2.5 Polynomial Model of Order 3

Table 6: Cubic Polynomial Model

Statistic	Value
Adjusted R^2	0.9839
Residual Standard Error σ	4.966
Cubic Term p -value	0.647

For the third-order polynomial model

$$Y = \beta_0 + \beta_1 X_c + \beta_2 X_c^2 + \beta_3 X_c^3 + \varepsilon,$$

we test whether the cubic term has an effect given that X_c and X_c^2 are already in the model.

$$H_0 : \beta_3 = 0 \quad \text{and} \quad H_1 : \beta_3 \neq 0.$$

The cubic term is not statistically significant since the p -value is 0.647, so we do not reject H_0 at the 5% level. This suggests that X_c^3 does not provide additional explanatory power once X_c and X_c^2 are included. The adjusted R^2 decreases slightly to 0.9839 and the residual standard error increases to 4.966, indicating that the extra complexity does not improve the model fit.

Therefore, the polynomial model of order 2 was selected. The regression summary and the final equation are presented in Table 7 and Equation (3).

Table 7: Summary of the selected model (Order 2)

Term	Estimate	Std. Error	<i>t</i> -statistic	<i>P</i> -value
Intercept	-30.74	0.4780	-64.31	$< 2 \times 10^{-16}$
X_c	0.3577	0.0090	39.82	$< 2 \times 10^{-16}$
X_c^2	0.0021	0.0001	29.79	$< 2 \times 10^{-16}$

$$\hat{Y}_2 = -30.74 + 0.3577X_c + 0.0021X_c^2 \quad (3)$$

5 Discussion

The figures below show the payoff/profit at expiration as a function of the terminal stock price S_T . Here we only look at the expiration outcome, so we do not calculate the put premium day by day.

5.1 Payoff at Maturity

At expiry, the stock payoff is simply S_T , so it increases as S_T increases. The put payoff is $\max(K - S_T, 0)$: it is positive when $S_T < K$, and it becomes 0 when $S_T \geq K$. So the stock line goes up, while the put line goes down to 0 and then stays at 0 after S_T reaches K (Figure 5).

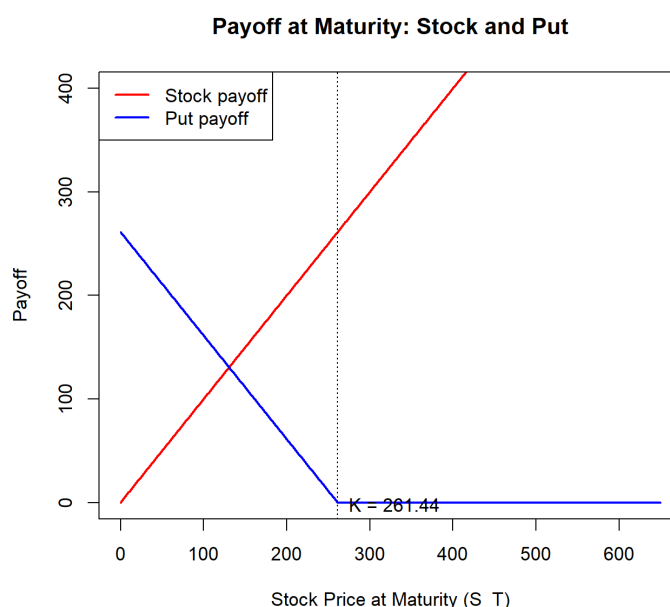


Figure 5: Payoff at Maturity: Long Stock and Long Put

5.2 Profit at Maturity

Figure 6 plots the expiry profit for a long stock and a long put. The stock profit is a straight increasing line as S_T increases. The put profit is positive when S_T is far below K , then it falls as S_T rises. Once $S_T \geq K$, the put finishes with zero payoff, so the profit is fixed at $-P_0 = -45.63$.

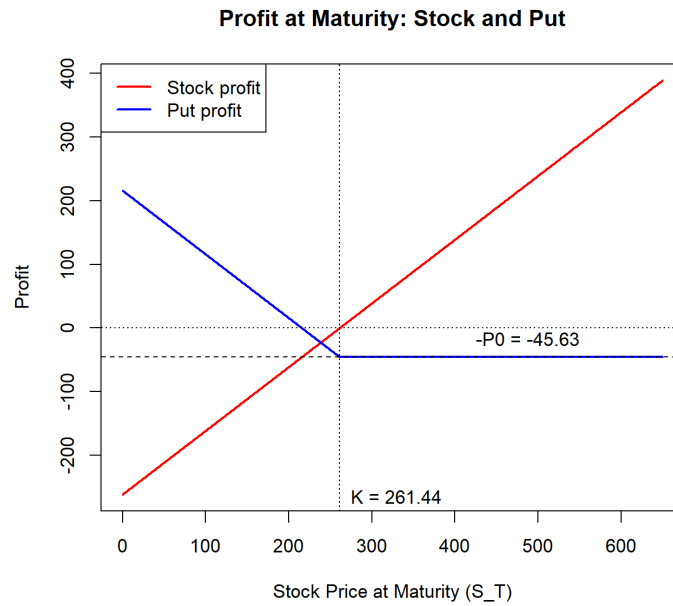


Figure 6: Profit at Maturity: Long Stock and Long Put

5.3 Protective Put (Stock + Put)

Figure 7 shows the protective put profit at expiry. The line is flat at -45.63 when S_T is below K , so the loss is capped at the premium paid. After S_T passes $K = 261.44$, the profit increases as S_T increases. The profit becomes zero at $S_T = 307.07$, which is the break-even point.

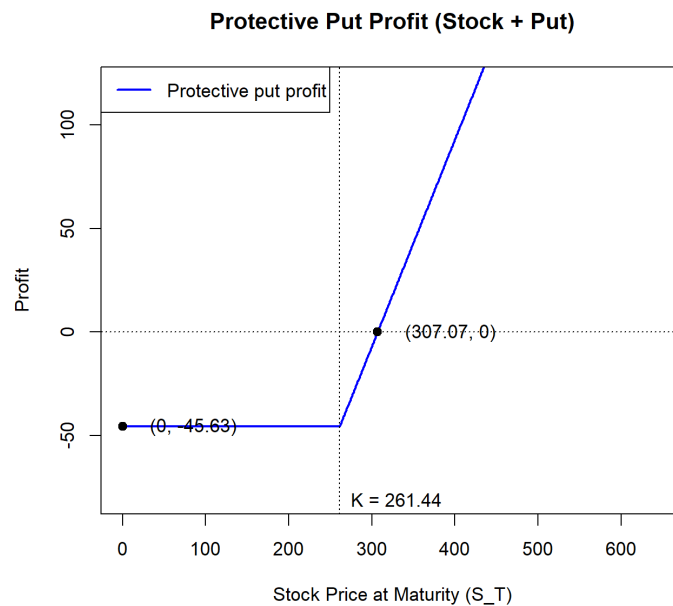


Figure 7: Protective Put Profit (Stock + Put)

6 Financial Interpretation

6.1 Breakeven Prices Analysis

We find the price X where profit is zero.

For the unhedged model, the profit equation is:

$$Y_1 = X - 261.44 \quad (4)$$

Setting $Y_1 = 0$ yields a breakeven price of:

$$X = 261.44 \text{ USD} \quad (5)$$

For the hedged model, the profit equation is:

$$Y_2 = -30.74 + 0.3577X_c + 0.0021X_c^2 \quad (6)$$

Setting $Y_2 = 0$, the root for the centered variable X_c is:

$$X_c \approx 62.81 \quad (7)$$

Given the mean $\bar{X} = 229.56$, the breakeven price X is:

$$X = 229.56 + 62.81 = 292.37 \text{ USD} \quad (8)$$

Table 8: Breakeven Prices for Hedged and Unhedged Strategies

Strategy	Breakeven Price in USD
Unhedged Strategy	261.44
Hedged Strategy	292.37

The unhedged strategy breaks even at 261.44 USD, while the hedged strategy breaks even at 292.37 USD. The hedged point is higher due to the option premium paid for the put option. This fee increases the total cost, so the stock price must rise further to cover the expense, indicating that risk reduction has a price.

6.2 Marginal Profit Analysis

We compare the marginal effect of the stock price on profit for the unhedged and hedged models.

- **Unhedged model (Y_1):**

$$\frac{dY_1}{dX} = 1.$$

The slope is constant. A 1 USD increase in X increases Y_1 by 1 USD, so the marginal effect does not depend on the price level.

- **Hedged model (Y_2):**

$$\frac{dY_2}{dX_c} = 0.3577 + 0.0042X_c.$$

The slope depends on X_c . Since the coefficient on X_c is positive, the marginal effect increases as X_c rises. Therefore, when the stock price is relatively low, changes in X_c have a smaller effect on Y_2 , while at higher prices the marginal increase in Y_2 is larger.

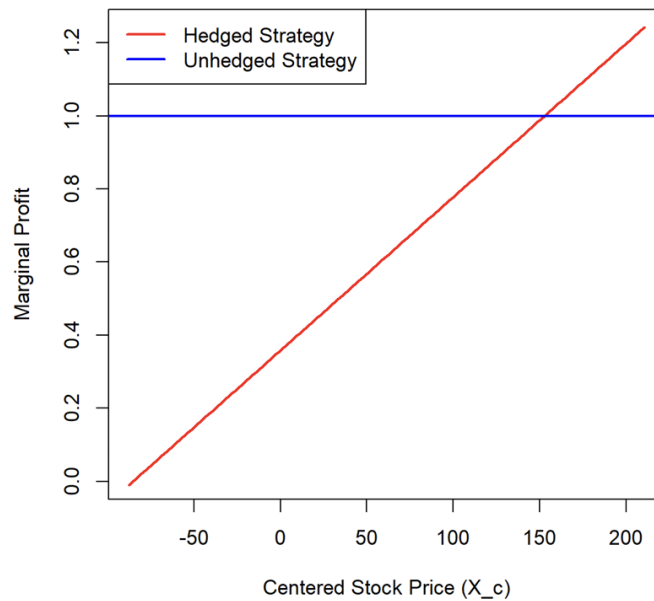


Figure 8: Marginal Profit Comparison Between Hedged and Unhedged Strategies

6.3 Downside Risk Comparison

Table 9: Risk characteristics of hedged and unhedged positions

Statistic (USD)	Unhedged Strategy	Hedged Strategy
Minimum Profit	−119.39	−49.91
Maximum Profit	178.69	133.06
Standard Deviation	65.88	39.14

Table 9 summarises the risk characteristics of the unhedged and hedged strategies in the training dataset. The unhedged strategy experiences a much lower minimum profit, showing larger losses during downturns. By contrast, the protective put limits the downside, so the minimum profit is less negative. The unhedged strategy achieves a higher maximum profit, but it also has a larger standard deviation, indicating greater volatility. Overall, the hedged strategy offers more stable outcomes at the cost of reduced upside.

6.4 Investor Profile Analysis

We identify the suitable investor for each strategy.

1. Unhedged Model This strategy suits aggressive investors. They expect the stock price to rise. They accept the risk of unlimited losses. Their goal is to maximize profit without paying for insurance costs.

2. Hedged Model This strategy suits conservative investors. They are concerned about price drops. They prioritize capital preservation. They are willing to pay the option premium to limit potential losses.

7 Validation and Conclusion

The model achieved high predictive accuracy on the 30% test set, as shown by points along the 45° line (Figure 9).

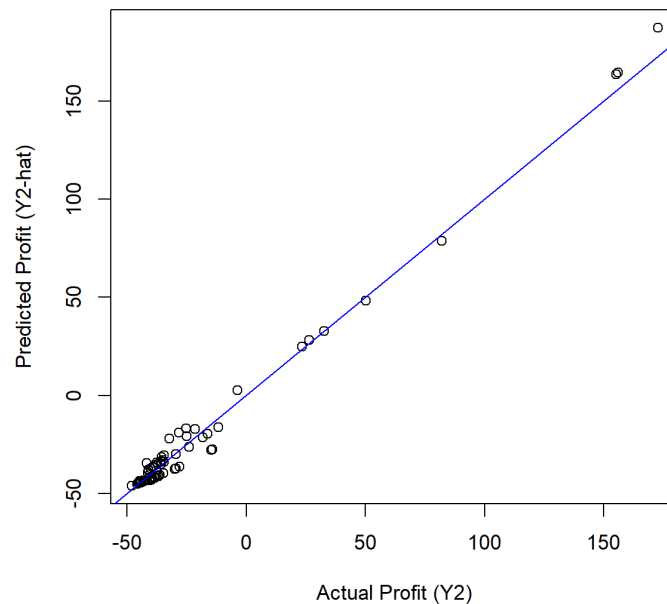


Figure 9: Predicted vs Actual Profit (Test Data)

Figure 10 contrasts the quadratic Hedged Profit (Red) with the linear Unhedged Profit (Blue). The polynomial fit captures the protective floor at low stock prices, illustrating the convexity that the linear model fails to represent.

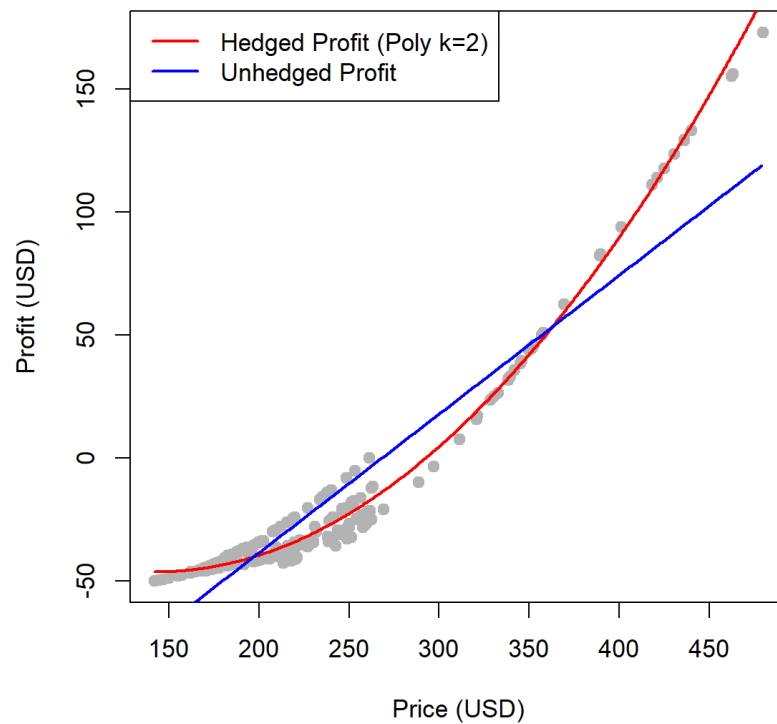


Figure 10: Model Comparison: Polynomial (Red) vs Linear (Blue)

The analysis shows that the quadratic model best fits the hedged strategy because the relationship between price and profit is non-linear. This strategy reduces losses to achieve effective risk management. Although paying for the option slightly lowers the profit when the stock price rises, it helps to control risk. This allows the investor to avoid large losses while still earning from future price increases.

References

- [1] Hull, J. C. (2021). *Options, futures, and other derivatives* (11th ed.). Pearson.
- [2] Montgomery, D. C., Peck, E. A., & Vining, G. G. (2012). *Introduction to linear regression analysis* (5th ed.). Wiley.
- [3] Yahoo Finance. (2025). *Tesla, Inc. (TSLA) historical data*. Retrieved October 27, 2025, from <https://finance.yahoo.com/quote/TSLA/history>

Appendix

A. R Code

```
1 data <- read.table(  
2   "C:/Users/Kow Chun Xiang/Desktop/fm2.txt",  
3   header = TRUE  
4 )  
5 data  
6  
7 set.seed(4)  
8 train_indices <- sample(nrow(data), nrow(data) * 0.7)  
9 data_e <- data[train_indices, ]  
10 data_p <- data[-train_indices, ]  
11  
12 plot(data_e$X, data_e$Y1,  
13       xlab="Tesla Stock Price (USD)",  
14       ylab="Unhedged profit (USD)",  
15       )  
16  
17 plot(data_e$X, data_e$Y2,  
18       xlab="Tesla Stock Price (USD)",  
19       ylab="Hedged profit (USD)",  
20       )  
21  
22 model_k1 <- lm(Y2 ~ X, data = data_e)  
23 summary(model_k1)  
24  
25 model_k2 <- lm(Y2 ~ X + I(X^2), data = data_e)  
26 summary(model_k2)  
27  
28 model_k3 <- lm(Y2 ~ X + I(X^2) + I(X^3), data = data_e)  
29 summary(model_k3)  
30  
31 library(car)  
32 data_e$X2 <- data_e$X^2  
33  
34 plot(  
35   data_e$X, data_e$X2,  
36   main = "Scatterplot: X vs X^2",  
37   xlab = "X",  
38   ylab = "X^2",  
39 )  
40  
41 cor(data_e$X, data_e$X^2)  
42
```

```

43 X_mean_e <- mean(data_e$X)
44 data_e$X_c <- data_e$X - X_mean_e
45 data_p$X_c <- data_p$X - X_mean_e
46
47 model_c1 <- lm(Y2 ~ X_c, data = data_e)
48 summary(model_c1)
49
50 model_c2 <- lm(Y2 ~ X_c + I(X_c^2), data = data_e)
51 summary(model_c2)
52
53 model_c3 <- lm(Y2 ~ X_c + I(X_c^2) + I(X_c^3), data = data_e)
54 summary(model_c3)
55
56 data_e$X2_c <- data_e$X_c^2
57 plot(data_e$X_c, data_e$X2_c,
58      main = "Scatterplot: Centered X vs Centered X^2",
59      xlab = "Centered X (X_c)",
60      ylab = "Centered X^2 (X2_c)",
61 )
62
63 cor(data_e$X_c, data_e$X_c^2)
64
65 t <- rstudent(model_c2)
66 qqnorm(t)
67 qqline(t, col = "red")
68
69 yhat <- model_c2$fit
70 plot(yhat, t)
71 abline(h = 0, col = "red")
72
73 ts.plot(t)
74 abline(h = 0, col = "red")
75
76 pred_fitted_c2 <- predict(model_c2, newdata = data_p)
77 plot(data_p$Y2, pred_fitted_c2,
78      xlab = "Actual Profit (Y2)",
79      ylab = "Predicted Profit (Y2-hat)",
80 )
81 abline(a = 0, b = 1, col = "blue")
82
83 model_slr <- lm(Y2 ~ X, data = data_e)
84
85 X_seq <- seq(min(data$X), max(data$X))
86 X_seq_c <- X_seq - X_mean_e
87 preds_poly <- predict(model_c2, newdata = data.frame(X_c = X_seq_c))
88 preds_slr <- predict(model_slr, newdata = data.frame(X = X_seq))
89

```

```

90 plot(data$X, data$Y2,
91       xlab = "Tesla Stock Price (USD)",
92       ylab = "Hedged Profit (USD)",
93       pch = 19, col = "gray70",
94       ylim = c(0, max(data$Y2) + 5))
95
96 lines(X_seq, preds_poly, col = "red", lwd = 2)
97 lines(X_seq, preds_slr, col = "blue", lwd = 2)
98
99 legend("topleft",
100       legend = c("Polynomial (k=2)", "Straight Line"),
101       col = c("red", "blue"),
102       lwd = 2)
103
104 summary(model_slr)
105 summary(model_c2)
106
107 mean(data_e$X)
108
109 library(car)
110 vif(model_k2)
111 vif(model_c2)
112
113
114 min(data_e$Y1)
115 min(data_e$Y2)
116
117 max(data_e$Y1)
118 max(data_e$Y2)
119
120 sd(data_e$Y1)
121 sd(data_e$Y2)
122
123 Xc_seq <- seq(min(data_e$X_c), max(data_e$X_c), length.out = 200)
124
125 marginal_unhedged <- rep(1, length(Xc_seq))
126 marginal_hedged <- 0.3577 + 0.0042 * Xc_seq
127
128 plot(Xc_seq, marginal_hedged,
129       type = "l", lwd = 2, col = "red",
130       xlab = "Centered Stock Price (X_c)",
131       ylab = "Marginal Profit",
132       ylim = c(0, max(marginal_unhedged, marginal_hedged)))
133
134 abline(h = 1, col = "blue", lwd = 2)
135
136 legend("topleft",

```

```

137     legend = c("Hedged Strategy", "Unhedged Strategy"),
138     col = c("blue", "red"),
139     lwd = 2)
140
141 X2 <- data_e$X^2
142
143 Xc2 <- data_e$X_c^2
144
145 par(mfrow = c(1, 2))
146
147 plot(data_e$X, X2,
148      xlab = "X",
149      ylab = "X^2",
150      main = "Before Mean Centering",
151      pch = 19, col = "gray")
152
153 plot(data_e$X_c, Xc2,
154      xlab = "Centered X (X_c)",
155      ylab = "Centered X^2",
156      main = "After Mean Centering",
157      pch = 19, col = "gray")
158
159 par(mfrow = c(1, 1))
160
161 options(scipen = 999)
162
163 data <- read.table(
164   "C:/Users/Kow Chun Xiang/Desktop/fm2.txt",
165   header = TRUE
166 )
167
168 bs_put <- function(S, K, r, sigma, T) {
169   d1 <- (log(S / K) + (r + 0.5 * sigma^2) * T) / (sigma * sqrt(T))
170   d2 <- d1 - sigma * sqrt(T)
171   K * exp(-r * T) * pnorm(-d2) - S * pnorm(-d1)
172 }
173
174 K <- 261.44
175 r <- 0.04
176 sigma <- 0.5
177
178 n <- nrow(data)
179
180 T_vec <- seq(n, 1, by = -1) / n
181
182 data$Put_Premium <- mapply(
183   bs_put,

```

```

184 S = data$X,
185 T = T_vec,
186 MoreArgs = list(K = K, r = r, sigma = sigma)
187 )
188
189 data$Put_Premium <- round(data$Put_Premium, 2)
190
191 print(data[, c("Seq", "Date", "X", "Put_Premium")])
192
193 data <- read.table(
194   "C:/Users/Kow Chun Xiang/Desktop/fm 2.txt", # <- change to your
        exact filename
195   header = TRUE
196 )
197
198 data$Date <- as.Date(data$Date) # format is YYYY-MM-DD, so this works
199
200 plot(data$Date, data$Put_Premium,
201       type = "l",
202       xlab = "Date",
203       ylab = "Put Premium (USD)",
204       main = "Put Premium Over Time")
205
206 points(data$Date, data$Put_Premium, pch = 16, cex = 0.4)
207 grid()
208
209 S0 <- 261.44
210 K <- 261.44
211 P0 <- 45.63
212
213 ST <- seq(0, 650, by = 1)
214
215 payoff_stock <- ST
216 profit_stock <- ST - S0
217
218 payoff_put <- pmax(K - ST, 0)
219 profit_put <- payoff_put - P0
220
221 payoff_pp <- payoff_stock + payoff_put
222 profit_pp <- profit_stock + profit_put
223
224 plot(ST, payoff_stock, type = "l", col = "red", lwd = 2,
225       xlab = "Stock Price at Maturity (S_T)", ylab = "Payoff",
226       main = "Payoff at Maturity: Stock and Put",
227       ylim = c(0, 400))
228 lines(ST, payoff_put, col = "blue", lwd = 2)
229 abline(v = K, lty = 3)

```

```

230
231 usr <- par("usr")
232 text(x = K, y = usr[3] + 0.03*(usr[4]-usr[3]),
233       labels = paste0("K = ", K), pos = 4, offset = 0.5)
234
235 legend("topleft",
236        legend = c("Stock payoff", "Put payoff"),
237        col = c("red", "blue"), lwd = 2, bty = "o")
238
239 plot(ST, profit_stock, type = "l", col = "red", lwd = 2,
240       xlab = "Stock Price at Maturity (S_T)", ylab = "Profit",
241       main = "Profit at Maturity: Stock and Put")
242 lines(ST, profit_put, col = "blue", lwd = 2)
243
244 abline(h = 0, lty = 3)
245 abline(v = K, lty = 3)
246 abline(h = -P0, lty = 2)
247
248 text(x = max(ST) * 0.75, y = -P0,
249       labels = paste0("-P0 = ", -1 * P0),
250       pos = 3)
251
252 usr <- par("usr")
253 text(x = K, y = usr[3] + 0.03*(usr[4]-usr[3]),
254       labels = paste0("K = ", K), pos = 4, offset = 0.5)
255
256 legend("topleft",
257        legend = c("Stock profit", "Put profit"),
258        col = c("red", "blue"),
259        lwd = c(2, 2),
260        lty = c(1, 1),
261        bty = "o")
262
263 plot(ST, profit_pp, type="l", col="blue", lwd=2,
264       xlab="Stock Price at Maturity (S_T)", ylab="Profit",
265       main="Protective Put Profit (Stock + Put)",
266       xlim=c(0, 650),
267       ylim=c(-80, 120))
268
269 abline(h = 0, lty = 3)
270 abline(v = K, lty = 3)
271
272 usr <- par("usr")
273 text(x = K, y = usr[3] + 0.03*(usr[4]-usr[3]),
274       labels = paste0("K = ", K), pos = 4, offset = 0.5)
275
276 points(0, -P0, pch = 19)

```

```

277 text(0, -P0, labels = paste0("(0, ", round(-P0, 2), ")"), pos = 4,
      offset = 1.2)
278
279 x0 <- K + P0
280 points(x0, 0, pch = 19)
281 text(x0, 0, labels = paste0("(", round(x0, 2), ", 0)"), pos = 4, offset
      = 1.2)
282
283 legend("topleft",
284        legend = c("Protective put profit"),
285        col = c("blue"),
286        lwd = 2,
287        bty = "o")

```

C. Python Code for Black-Scholes Put Option Pricing

```

1 import math
2 from scipy.stats import norm
3
4 def calculate_put_option_price():
5     """
6     Calculate Black-Scholes Put Option Price.
7     """
8     print("=== Black-Scholes Model Calculator ===")
9
10    try:
11        S = float(input("Enter Stock Price (S): "))
12        K = float(input("Enter Strike Price (K): "))
13        r = float(input("Enter Risk-free Rate (r): "))
14        T = float(input("Enter Time to Maturity (T): "))
15        sigma = float(input("Enter Volatility (sigma): "))
16
17        if T <= 0 or sigma <= 0 or S <= 0 or K <= 0:
18            print("Error: All inputs must be positive.")
19            return
20
21        d1 = (math.log(S / K) + (r + (sigma ** 2) / 2) * T) / (sigma *
math.sqrt(T))
22        d2 = d1 - sigma * math.sqrt(T)
23
24        N_minus_d1 = norm.cdf(-d1)
25        N_minus_d2 = norm.cdf(-d2)
26
27        P = K * math.exp(-r * T) * N_minus_d2 - S * N_minus_d1
28

```



```
29     print("Put Option Price (P) =", round(P, 4))
30
31     except ValueError:
32         print("Error: Please enter valid numbers.")
33     except Exception as e:
34         print("An error occurred:", e)
35
36 if __name__ == "__main__":
37     calculate_put_option_price()
```

B. Full Dataset

Seq	Date	Y1	Y2	X	Put_Premium						
						56	2024-03-18	-87.64	-43.69	173.80	89.59
1	2023-12-27	0.00	0.00	261.44	45.63	57	2024-03-19	-90.12	-44.36	171.32	91.40
2	2023-12-28	-8.26	-5.17	253.18	48.72	58	2024-03-20	-85.78	-43.26	175.66	88.16
3	2023-12-29	-12.96	-8.05	248.48	50.54	59	2024-03-21	-88.62	-44.03	172.82	90.23
4	2024-01-02	-13.02	-8.16	248.42	50.49	60	2024-03-22	-90.61	-44.56	170.83	91.69
5	2024-01-03	-22.99	-13.94	238.45	54.68	61	2024-03-25	-88.81	-44.14	172.63	90.31
6	2024-01-04	-23.51	-14.30	237.93	54.84	62	2024-03-26	-83.77	-42.84	177.67	86.57
7	2024-01-05	-23.95	-14.61	237.49	54.97	63	2024-03-27	-81.61	-42.28	179.83	84.98
8	2024-01-08	-20.99	-13.03	240.45	53.59	64	2024-03-28	-85.65	-43.41	175.79	87.88
9	2024-01-09	-26.48	-16.14	234.96	55.98	65	2024-04-01	-86.22	-43.59	175.22	88.27
10	2024-01-10	-27.50	-16.76	233.94	56.38	66	2024-04-02	-94.81	-45.69	166.63	94.76
11	2024-01-11	-34.22	-20.36	227.22	59.49	67	2024-04-03	-93.06	-45.32	168.38	93.38
12	2024-01-12	-42.55	-24.59	218.89	63.60	68	2024-04-04	-90.33	-44.70	171.11	91.26
13	2024-01-16	-41.53	-24.15	219.91	63.01	69	2024-04-05	-96.54	-46.13	164.90	96.05
14	2024-01-17	-45.89	-26.30	215.55	65.23	70	2024-04-08	-88.46	-44.31	172.98	89.79
15	2024-01-18	-49.56	-28.06	211.88	67.14	71	2024-04-09	-84.56	-43.35	176.88	86.85
16	2024-01-19	-49.25	-27.97	212.19	66.91	72	2024-04-10	-89.68	-44.66	171.76	90.66
17	2024-01-22	-52.64	-29.55	208.80	68.72	73	2024-04-11	-86.84	-44.00	174.60	88.49
18	2024-01-23	-52.30	-29.46	209.14	68.48	74	2024-04-12	-90.39	-44.88	171.05	91.15
19	2024-01-24	-53.61	-30.09	207.83	69.16	75	2024-04-15	-99.96	-46.93	161.48	98.67
20	2024-01-25	-78.81	-39.86	182.63	84.59	76	2024-04-16	-104.33	-47.74	157.11	102.23
21	2024-01-26	-78.19	-39.69	183.25	84.14	77	2024-04-17	-105.99	-48.03	155.45	103.60
22	2024-01-29	-70.51	-37.03	190.93	79.11	78	2024-04-18	-111.51	-48.89	149.93	108.25
23	2024-01-30	-69.85	-36.83	191.59	78.65	79	2024-04-19	-114.39	-49.29	147.05	110.73
24	2024-01-31	-74.15	-38.42	187.29	81.37	80	2024-04-22	-119.39	-49.91	142.05	115.12
25	2024-02-01	-72.58	-37.91	188.86	80.31	81	2024-04-23	-116.76	-49.59	144.68	112.80
26	2024-02-02	-73.53	-38.28	187.91	80.88	82	2024-04-24	-99.31	-46.92	162.13	98.03
27	2024-02-05	-80.38	-40.62	181.06	85.39	83	2024-04-25	-91.26	-45.31	170.18	91.59
28	2024-02-06	-76.34	-39.33	185.10	82.65	84	2024-04-26	-93.15	-45.74	168.29	93.05
29	2024-02-07	-73.86	-38.52	187.58	80.98	85	2024-04-29	-67.39	-38.95	194.05	74.09
30	2024-02-08	-71.88	-37.87	189.56	79.65	86	2024-04-30	-78.16	-42.20	183.28	81.61
31	2024-02-09	-67.87	-36.46	193.57	77.05	87	2024-05-01	-81.45	-43.10	179.99	83.99
32	2024-02-12	-73.31	-38.45	188.13	80.49	88	2024-05-02	-81.43	-43.14	180.01	83.94
33	2024-02-13	-77.42	-39.88	184.02	83.18	89	2024-05-03	-80.25	-42.87	181.19	83.03
34	2024-02-14	-72.73	-38.34	188.71	80.03	90	2024-05-06	-76.68	-41.96	184.76	80.37
35	2024-02-15	-60.99	-34.02	200.45	72.61	91	2024-05-07	-83.63	-43.79	177.81	85.49
36	2024-02-16	-61.49	-34.27	199.95	72.86	92	2024-05-08	-86.72	-44.55	174.72	87.82
37	2024-02-20	-67.68	-36.67	193.76	76.65	93	2024-05-09	-89.47	-45.18	171.97	89.93
38	2024-02-21	-66.67	-36.35	194.77	75.96	94	2024-05-10	-92.97	-45.93	168.47	92.68
39	2024-02-22	-64.03	-35.41	197.41	74.26	95	2024-05-13	-89.55	-45.25	171.89	89.94
40	2024-02-23	-69.47	-37.45	191.97	77.66	96	2024-05-14	-83.89	-44.02	177.55	85.51
41	2024-02-26	-62.04	-34.75	199.40	72.94	97	2024-05-15	-87.45	-44.86	173.99	88.23
42	2024-02-27	-61.71	-34.67	199.73	72.68	98	2024-05-16	-86.60	-44.70	174.84	87.54
43	2024-02-28	-59.40	-33.82	202.04	71.22	99	2024-05-17	-83.98	-44.14	177.46	85.48
44	2024-02-29	-59.56	-33.94	201.88	71.27	100	2024-05-20	-86.49	-44.74	174.95	87.40
45	2024-03-01	-58.80	-33.69	202.64	70.75	101	2024-05-21	-74.84	-41.94	186.60	78.55
46	2024-03-04	-73.30	-39.04	188.14	79.90	102	2024-05-22	-81.33	-43.63	180.11	83.35
47	2024-03-05	-80.70	-41.44	180.74	84.90	103	2024-05-23	-87.70	-45.09	173.74	88.26
48	2024-03-06	-84.90	-42.70	176.54	87.84	104	2024-05-24	-82.20	-43.91	179.24	83.94
49	2024-03-07	-82.79	-42.13	178.65	86.30	105	2024-05-28	-84.69	-44.50	176.75	85.83
50	2024-03-08	-86.10	-43.10	175.34	88.64	106	2024-05-29	-85.25	-44.66	176.19	86.24
51	2024-03-11	-83.67	-42.45	177.77	86.86	107	2024-05-30	-82.65	-44.12	178.79	84.18
52	2024-03-12	-83.90	-42.54	177.54	86.99	108	2024-05-31	-83.36	-44.31	178.08	84.70
53	2024-03-13	-91.96	-44.71	169.48	92.88	109	2024-06-03	-85.15	-44.73	176.29	86.06
54	2024-03-14	-98.94	-46.36	162.50	98.21	110	2024-06-04	-86.67	-45.08	174.77	87.24
55	2024-03-15	-97.87	-46.15	163.57	97.36	111	2024-06-05	-86.44	-45.06	175.00	87.02

112	2024-06-06	-83.50	-44.48	177.94	84.67	171	2024-08-30	-47.33	-38.87	214.11	54.16
113	2024-06-07	-83.96	-44.61	177.48	85.00	172	2024-09-03	-50.84	-39.98	210.60	56.55
114	2024-06-10	-87.65	-45.39	173.79	87.90	173	2024-09-04	-42.03	-37.41	219.41	50.33
115	2024-06-11	-90.78	-46.01	170.66	90.42	174	2024-09-05	-31.27	-33.65	230.17	43.34
116	2024-06-12	-84.15	-44.75	177.29	85.05	175	2024-09-06	-50.71	-40.23	210.73	56.18
117	2024-06-13	-78.97	-43.65	182.47	80.97	176	2024-09-09	-45.17	-38.73	216.27	52.15
118	2024-06-14	-83.43	-44.67	178.01	84.41	177	2024-09-10	-35.27	-35.54	226.17	45.45
119	2024-06-17	-74.00	-42.55	187.44	77.10	178	2024-09-11	-33.31	-34.95	228.13	44.08
120	2024-06-18	-76.58	-43.22	184.86	79.01	179	2024-09-12	-31.63	-34.45	229.81	42.91
121	2024-06-20	-79.87	-44.01	181.57	81.51	180	2024-09-13	-31.15	-34.40	230.29	42.48
122	2024-06-21	-78.43	-43.73	183.01	80.35	181	2024-09-16	-34.66	-35.83	226.78	44.55
123	2024-06-24	-78.86	-43.87	182.58	80.64	182	2024-09-17	-33.57	-35.57	227.87	43.73
124	2024-06-25	-74.09	-42.81	187.35	76.94	183	2024-09-18	-34.24	-35.95	227.20	44.02
125	2024-06-26	-65.07	-40.49	196.37	70.24	184	2024-09-19	-17.52	-29.25	243.92	34.02
126	2024-06-27	-64.02	-40.25	197.42	69.43	185	2024-09-20	-23.19	-31.92	238.25	37.02
127	2024-06-28	-63.56	-40.18	197.88	69.04	186	2024-09-23	-11.44	-26.67	250.00	30.53
128	2024-07-01	-51.58	-36.45	209.86	60.80	187	2024-09-24	-7.17	-24.67	254.27	28.27
129	2024-07-02	-30.18	-28.00	231.26	47.87	188	2024-09-25	-4.42	-23.38	257.02	26.80
130	2024-07-03	-15.05	-20.72	246.39	40.02	189	2024-09-26	-7.22	-25.05	254.22	27.94
131	2024-07-05	-9.92	-18.10	251.52	37.52	190	2024-09-27	-0.98	-21.86	260.46	24.89
132	2024-07-08	-8.50	-17.44	252.94	36.76	191	2024-09-30	0.19	-21.39	261.63	24.19
133	2024-07-09	0.89	-12.18	262.33	32.63	192	2024-10-01	-3.42	-23.58	258.02	25.62
134	2024-07-10	1.82	-11.75	263.26	32.13	193	2024-10-02	-12.42	-28.38	249.02	29.81
135	2024-07-11	-20.41	-24.03	241.03	42.07	194	2024-10-03	-20.78	-32.35	240.66	34.19
136	2024-07-12	-13.21	-20.47	248.23	38.44	195	2024-10-04	-11.36	-28.23	250.08	28.90
137	2024-07-15	-8.80	-18.21	252.64	36.28	196	2024-10-07	-20.61	-32.62	240.83	33.75
138	2024-07-16	-4.88	-16.16	256.56	34.43	197	2024-10-08	-16.94	-31.21	244.50	31.51
139	2024-07-17	-12.94	-20.69	248.50	37.95	198	2024-10-09	-20.39	-32.88	241.05	33.28
140	2024-07-18	-12.21	-20.43	249.23	37.48	199	2024-10-10	-22.67	-33.99	238.77	34.45
141	2024-07-19	-22.24	-25.62	239.20	42.32	200	2024-10-11	-43.64	-41.01	217.80	48.36
142	2024-07-22	-9.93	-19.46	251.51	36.18	201	2024-10-14	-42.28	-40.79	219.16	47.23
143	2024-07-23	-15.06	-22.29	246.38	38.47	202	2024-10-15	-41.87	-40.81	219.57	46.80
144	2024-07-24	-45.45	-35.66	215.99	55.47	203	2024-10-16	-40.11	-40.47	221.33	45.38
145	2024-07-25	-41.19	-34.19	220.25	52.69	204	2024-10-17	-40.55	-40.72	220.89	45.57
146	2024-07-26	-41.64	-34.46	219.80	52.87	205	2024-10-18	-40.74	-40.90	220.70	45.58
147	2024-07-29	-29.34	-29.57	232.10	45.46	206	2024-10-21	-42.59	-41.50	218.85	46.83
148	2024-07-30	-38.82	-33.59	222.62	50.92	207	2024-10-22	-43.47	-41.83	217.97	47.37
149	2024-07-31	-29.37	-29.82	232.07	45.25	208	2024-10-23	-47.79	-42.89	213.65	50.63
150	2024-08-01	-44.58	-35.92	216.86	54.35	209	2024-10-24	-0.96	-25.76	260.48	21.02
151	2024-08-02	-53.77	-39.01	207.67	60.44	210	2024-10-25	7.75	-20.95	269.19	17.12
152	2024-08-05	-62.56	-41.53	198.88	66.71	211	2024-10-28	1.07	-25.10	262.51	19.66
153	2024-08-06	-60.80	-41.14	200.64	65.33	212	2024-10-29	-1.92	-26.98	259.52	20.77
154	2024-08-07	-69.68	-43.32	191.76	72.03	213	2024-10-30	-3.89	-28.26	257.55	21.46
155	2024-08-08	-62.60	-41.74	198.84	66.54	214	2024-10-31	-11.59	-32.26	249.85	25.15
156	2024-08-09	-61.44	-41.52	200.00	65.60	215	2024-11-01	-12.46	-32.89	248.98	25.40
157	2024-08-12	-63.95	-42.20	197.49	67.42	216	2024-11-04	-18.60	-35.70	242.84	28.72
158	2024-08-13	-53.61	-39.54	207.83	59.75	217	2024-11-05	-10.00	-32.25	251.44	23.60
159	2024-08-14	-60.06	-41.38	201.38	64.36	218	2024-11-06	27.09	-9.78	288.53	8.97
160	2024-08-15	-47.30	-37.79	214.14	55.20	219	2024-11-07	35.47	-3.54	296.91	6.81
161	2024-08-16	-45.32	-37.24	216.12	53.78	220	2024-11-08	59.78	17.01	321.22	3.00
162	2024-08-19	-38.72	-35.04	222.72	49.38	221	2024-11-11	88.56	43.85	350.00	0.99
163	2024-08-20	-40.34	-35.73	221.10	50.31	222	2024-11-12	67.05	23.37	328.49	2.07
164	2024-08-21	-38.17	-35.06	223.27	48.81	223	2024-11-13	68.80	24.87	330.24	1.82
165	2024-08-22	-50.78	-39.32	210.66	57.15	224	2024-11-14	49.74	7.40	311.18	3.45
166	2024-08-23	-41.12	-36.33	220.32	50.49	225	2024-11-15	59.28	15.79	320.72	2.28
167	2024-08-26	-48.23	-38.75	213.21	55.18	226	2024-11-18	77.30	32.60	338.74	1.02
168	2024-08-27	-52.23	-40.01	209.21	57.91	227	2024-11-19	84.56	39.54	346.00	0.67
169	2024-08-28	-55.69	-41.03	205.75	60.35	228	2024-11-20	80.59	35.61	342.03	0.72
170	2024-08-29	-55.16	-40.97	206.28	59.88	229	2024-11-21	78.20	33.22	339.64	0.72

230	2024-11-22	91.12	45.78	352.56	0.34	241	2024-12-10	139.55	93.92	400.99	0.00
231	2024-11-25	77.15	32.04	338.59	0.60	242	2024-12-11	163.33	117.70	424.77	0.00
232	2024-11-26	76.79	31.62	338.23	0.53	243	2024-12-12	156.66	111.03	418.10	0.00
233	2024-11-27	71.45	26.35	332.89	0.61	244	2024-12-13	174.79	129.16	436.23	0.00
234	2024-11-29	83.72	38.31	345.16	0.26	245	2024-12-16	201.58	155.95	463.02	0.00
235	2024-12-02	95.65	50.10	357.09	0.10	246	2024-12-17	218.42	172.79	479.86	0.00
236	2024-12-03	89.98	44.44	351.42	0.12	247	2024-12-18	178.69	133.06	440.13	0.00
237	2024-12-04	96.49	50.90	357.93	0.06	248	2024-12-19	174.73	129.10	436.17	0.00
238	2024-12-05	108.05	62.43	369.49	0.02	249	2024-12-20	159.62	113.99	421.06	0.00
239	2024-12-06	127.78	82.15	389.22	0.00	250	2024-12-23	169.16	123.53	430.60	0.00
240	2024-12-09	128.35	82.72	389.79	0.00	251	2024-12-24	200.84	155.21	462.28	0.00